

Multiple Factor Analysis

Analysis 1: Combined Impact of Study Hours and Attendance on Exam Performance

Objective

To examine how the combination of study hours and attendance influences students' academic performance.

Hypothesis

Students who study more and maintain excellent attendance are expected to achieve the best academic performance.

Results

The analysis revealed a strong combined effect of study hours and attendance on academic performance. Students with **High Study Hours** and **Excellent Attendance** achieved the highest average exam score (**71.44**) and the smallest average decline in performance (**-3.86**). In contrast, students with **Low Study Hours** and **Poor Attendance** recorded the lowest average exam score (**61.42**) and the greatest decline (**-16.39**).

Students with high study hours but poor attendance performed noticeably worse than those with excellent attendance, indicating that regular attendance enhances the benefits of studying.

Conclusion

The combination of high study hours and excellent attendance produced the best academic outcomes. The results suggest that these two factors complement each other, and improving only one of them cannot fully compensate for poor performance in the other.

	AZ Study_Level	AZ Attendance_Level	123 Students	123 Avg_Exam	123 Avg_Improvement
1	High	Excellent	927	71.44	-3.86
2	Medium	Excellent	746	68.85	-5.04
3	High	Good	1,776	68.29	-6.77
4	Low	Excellent	72	66.35	-6.33
5	Medium	Good	1,377	66	-8.92
6	High	Poor	824	65.56	-10.59
7	Low	Good	135	63.61	-13.01
8	Medium	Poor	685	62.85	-11.89
9	Low	Poor	64	61.42	-16.39

Analysis 2: Combined Impact of Study Hours and Motivation Level on Exam Performance

Objective

To examine whether motivation level enhances the effect of study hours on students' academic performance.

Hypothesis

Students with both high motivation and longer study hours are expected to achieve the highest academic performance.

Results

The analysis showed that **High Motivation** consistently led to better academic performance across all study levels. Students with **High Motivation** and **High Study Hours** achieved the highest average exam score (**68.92**), while students with **Low Motivation** and **Low Study Hours** recorded the lowest average score (**63.01**) and the greatest decline in performance (**-13.60**).

Interestingly, highly motivated students maintained strong performance even with medium or low study hours, outperforming students with low motivation who studied the same amount.

Conclusion

The results indicate that motivation significantly enhances the benefits of studying. Students with high motivation consistently performed better regardless of study level, while the combination of low motivation and limited study hours produced the weakest academic outcomes.

	AZ motivation_level	AZ Study_Level	123 Students	123 Avg_Exam	123 Avg_Improvement
1	High	High	682	68.92	-6.65
2	Medium	High	1,795	68.57	-6.39
3	Low	High	1,050	68.05	-7.92
4	High	Medium	584	66.51	-9.26
5	Medium	Medium	1,415	66.12	-8.32
6	Low	Medium	809	65.39	-8.66
7	High	Low	53	65.23	-8.47
8	Medium	Low	141	63.73	-12.52
9	Low	Low	77	63.01	-13.6

Analysis 3: Combined Impact of Family Income and Access to Educational Resources on Exam Performance

Objective

To investigate whether access to educational resources can improve academic performance across different family income levels.

Hypothesis

Students with greater access to educational resources are expected to achieve better academic performance regardless of their family income.

Results

The analysis showed that students with **High Family Income** and **High Access to Educational Resources** achieved the highest average exam score (**68.61**) and maintained strong academic performance (**-6.84**). Conversely, students with **Medium Family Income** and **Low Access to Resources** recorded the lowest average exam score (**65.91**).

Across all income levels, students with **High Access to Resources** consistently outperformed those with **Medium** or **Low Access**, indicating that educational resources positively influence academic achievement regardless of family income.

Conclusion

The results suggest that access to educational resources plays a significant role in student success. While higher family income is associated with better performance, providing greater educational resources appears to improve outcomes for students across all income groups and may help reduce the academic gap between different income levels.

	AZ income_level	AZ resource_level	123 Students	123 Avg_Exam	123 Avg_Improvement
1	High	High	381	68.61	-6.84
2	Medium	High	763	68.51	-7.61
3	High	Medium	649	67.72	-6.57
4	Low	High	831	67.47	-7.2
5	Medium	Medium	1,361	67.24	-7.85
6	High	Low	238	66.82	-7.18
7	Low	Medium	1,308	66.71	-8.99
8	Low	Low	533	66.23	-8.62
9	Medium	Low	542	65.91	-8.06

Analysis 4: Combined Impact of Parental Education Level and Parental Involvement on Exam Performance

Objective

To examine how parental education level and parental involvement together influence students' academic performance.

Hypothesis

Students whose parents have higher education levels and are actively involved in their education are expected to achieve the best academic performance.

Results

The analysis showed that students with **Postgraduate-Educated Parents** and **High Parental Involvement** achieved the highest average exam score (**68.68**) and the smallest decline in performance (**-5.23**). In contrast, students whose parents had a **High School** education and **Low Parental Involvement** recorded the lowest average exam score (**65.93**) and one of the largest performance declines (**-9.04**).

Across all parental education levels, students with **High Parental Involvement** consistently outperformed those with **Medium** or **Low** involvement, indicating that active parental support has a strong positive effect regardless of parents' educational background.

Conclusion

The results suggest that parental involvement has a greater impact on students' academic performance than parental education level alone. While higher parental education contributes to better outcomes, active parental involvement consistently enhances student performance across all education levels.

	A2 education_level	A2 involvement_name	123 Students	123 Avg_Exam	123 Avg_Improvement
1	Postgraduate	High	365	68.68	-5.23
2	College	High	594	68.22	-6.32
3	Postgraduate	Medium	671	67.93	-6.98
4	High School	High	949	67.79	-7.42
5	College	Medium	981	67.13	-7.1
6	Postgraduate	Low	269	67.1	-9.98
7	High School	Medium	1,710	66.75	-8.75
8	College	Low	414	66.47	-9.31
9	High School	Low	653	65.93	-9.04

Analysis 5: Combined Impact of Teacher Quality and Family Income on Exam Performance

Objective

To investigate whether high teacher quality can reduce the academic performance gap associated with different family income levels.

Hypothesis

Students taught by high-quality teachers are expected to perform better regardless of their family income.

Results

The analysis showed that students with **High Teacher Quality** and **High Family Income** achieved the highest average exam score (**68.21**) and the smallest decline in performance (**-6.11**). In contrast, students with **Low Teacher Quality** and **Low Family Income** recorded the lowest average exam score (**66.42**) and the largest decline (**-11.00**).

Across all income levels, students taught by **High-Quality Teachers** consistently achieved higher average exam scores than those taught by medium- or low-quality teachers, suggesting that teacher quality positively influences academic performance regardless of family income.

Conclusion

The results indicate that teacher quality plays an important role in supporting students' academic achievement across all family income levels. While higher family income contributes to better performance, high-quality teaching consistently improves academic outcomes and may help reduce the negative effects associated with lower family income.

	AZ quality_level	AZ income_level	123 Students	123 Avg_Exam	123 Avg_Improvement
1	High	High	360	68.21	-6.11
2	High	Medium	785	67.73	-7.97
3	Medium	High	776	67.67	-6.69
4	Low	High	132	67.6	-8.98
5	High	Low	801	67.35	-8.05
6	Medium	Medium	1,639	67.24	-7.76
7	Low	Medium	242	66.68	-7.75
8	Medium	Low	1,588	66.67	-8.04
9	Low	Low	283	66.42	-11

Analysis 6: Combined Impact of Teacher Quality and Access to Educational Resources on Exam Performance

Objective

To investigate how teacher quality and access to educational resources together influence students' academic performance.

Hypothesis

Students with both high-quality teachers and high access to educational resources are expected to achieve the best academic performance.

Results

The analysis showed that students with **High Teacher Quality** and **High Access to Educational Resources** achieved the highest average exam score (**68.45**). Conversely, students with **Low Teacher Quality** and **Low Access to Resources** recorded the lowest average exam score (**65.78**) and the largest decline in performance (**-10.41**).

Across all teacher quality levels, students with **High Access to Resources** consistently outperformed those with **Medium** or **Low Access**, highlighting the complementary effect of these two factors.

Conclusion

The results indicate that teacher quality and access to educational resources work together to improve academic performance. Students achieved the best outcomes when both factors were at high levels, while limited resources combined with low teacher quality produced the weakest academic results.

	A2 quality_level	A2 resource_level	I23 Students	I23 Avg_Exam	I23 Avg_Improvement
1	High	High	581	68.45	-7.52
2	Medium	High	1,187	68.05	-6.82
3	High	Medium	957	67.56	-7.75
4	Low	High	207	67.34	-9.3
5	Medium	Medium	2,045	66.97	-8.04
6	Low	Medium	316	66.78	-9.03
7	High	Low	408	66.77	-7.64
8	Medium	Low	771	65.98	-7.98
9	Low	Low	134	65.78	-10.41

Analysis 7: Combined Impact of Peer Influence and Motivation Level on Exam Performance

Objective

To examine how peer influence and motivation level together affect students' academic performance.

Hypothesis

Students with positive peer influence and high motivation are expected to achieve the highest academic performance.

Results

The analysis showed that students with **Positive Peer Influence** and **High Motivation** achieved the highest average exam score (**68.01**). In contrast, students with **Negative Peer Influence** and **Low Motivation** recorded the lowest average exam score (**66.31**) and experienced a greater decline in performance.

Across all motivation levels, students with **Positive Peer Influence** consistently achieved higher exam scores than those with **Neutral** or **Negative** peer influence, indicating that a supportive social environment enhances academic performance.

Conclusion

The results suggest that peer influence and motivation complement each other in improving academic performance. Students with positive peer environments consistently performed better, while the combination of negative peer influence and low motivation was associated with the weakest academic outcomes.

	AZ influence_name	AZ motivation_level	123 Students	123 Avg_Exam	123 Avg_Improvement
1	Positive	High	533	68.01	-7.14
2	Positive	Medium	1,322	67.79	-6.43
3	Neutral	High	500	67.66	-7.96
4	Neutral	Medium	1,355	67.32	-7.88
5	Negative	High	286	67.21	-9.12
6	Positive	Low	782	67.04	-8.04
7	Neutral	Low	737	66.66	-8.92
8	Negative	Medium	674	66.45	-8.64
9	Negative	Low	417	66.31	-8.42

Analysis 8: Combined Impact of Previous Academic Performance and Study Hours on Exam Performance

Objective

To investigate whether increasing study hours can help students with different levels of previous academic performance improve their current exam results.

Hypothesis

Students with lower previous academic performance are expected to benefit more from increased study hours than students who previously performed well.

Results

The analysis showed that students with **Low Previous Scores** experienced the greatest improvement in performance. Those who studied **High** hours improved by an average of **12.97** points, compared with **10.42** and **8.63** points for students with medium and low study hours, respectively.

Students with **Medium Previous Scores** showed only a slight decline in performance, with higher study hours helping to minimize the decrease. Students with **High Previous Scores** achieved the highest average exam score (**69.16**) when studying for many hours, although they experienced the largest decline compared with their previous results due to their already high baseline performance.

Conclusion

The results suggest that increased study hours are particularly beneficial for students with lower previous academic performance, enabling them to achieve substantial improvements. While students with high previous scores maintained the highest overall exam scores, they had less room for improvement, making performance gains more difficult to achieve.

	AZ Previous_Level	AZ Study_Level	123 Students	123 Avg_Exam	123 Avg_Improvement
1	Low	High	643	67.73	12.97
2	Low	Medium	543	65.08	10.42
3	Low	Low	49	63.41	8.63
4	Medium	High	1,409	68.11	-1.52
5	Medium	Medium	1,142	65.78	-3.54
6	Medium	Low	106	63.56	-6.49
7	High	High	1,475	69.16	-20.69
8	High	Medium	1,123	66.65	-22.98
9	High	Low	116	64.23	-25.83

Analysis 9: Combined Impact of Learning Disabilities and Teacher Quality on Exam Performance

Objective

To investigate whether high teacher quality can help reduce the academic performance gap for students with learning disabilities.

Hypothesis

Students with learning disabilities are expected to perform better when taught by high-quality teachers.

Results

The analysis showed that students **without learning disabilities** and **High Teacher Quality** achieved the highest average exam score (**67.74**) and the smallest decline in performance (**-7.34**). In contrast, students **with learning disabilities** and **Low Teacher Quality** recorded the lowest average exam score (**65.43**) and the greatest decline (**-11.50**).

For students with learning disabilities, higher teacher quality was associated with better academic outcomes. Their average exam score increased from **65.43** with low-quality teachers to **66.95** with high-quality teachers, while the average decline in performance decreased from **-11.50** to **-10.42**.

Conclusion

The results indicate that teacher quality can partially reduce the negative impact of learning disabilities on academic performance. Although students with learning disabilities continued to score lower than their peers, those taught by high-quality teachers consistently achieved better outcomes than those taught by medium- or low-quality teachers.

	AZ disability_name	AZ quality_level	123 Students	123 Avg_Exam	123 Avg_Improvement
1	No	High	1,743	67.74	-7.34
2	No	Medium	3,579	67.22	-7.62
3	Yes	High	203	66.95	-10.42
4	No	Low	589	66.91	-9.15
5	Yes	Medium	424	66.08	-8.08
6	Yes	Low	68	65.43	-11.5

Analysis 10: Combined Impact of Distance from Home and Attendance on Exam Performance

Objective

To investigate whether attendance can reduce the impact of living farther from school on students' academic performance.

Hypothesis

Students with excellent attendance are expected to achieve strong academic performance regardless of their distance from school.

Results

The analysis showed that students with **Excellent Attendance** consistently achieved the highest exam scores across all distance categories. Students living **Near** the school with **Excellent Attendance** recorded the highest average exam score (**70.39**) and the smallest decline in performance (**-4.00**). In contrast, students with **Poor Attendance** and living **Far** from school achieved the lowest average exam score (**63.41**) and experienced the greatest decline (**-11.89**).

Across all distance categories, attendance had a much greater impact on academic performance than commuting distance. Students with excellent attendance consistently outperformed those with good or poor attendance, regardless of where they lived.

Conclusion

The results suggest that attendance is a stronger predictor of academic performance than distance from home. Although students living closer to school achieved slightly better results, maintaining excellent attendance had the greatest positive impact on academic achievement, regardless of commuting distance.

	AZ distance_level	AZ Attendance_Level	123 Students	123 Avg_Exam	123 Avg_Improvement
1	Near	Excellent	1,069	70.39	-4
2	Moderate	Excellent	525	69.74	-5.14
3	Far	Excellent	151	69.58	-5.42
4	Near	Good	1,957	67.34	-7.91
5	Moderate	Good	982	66.97	-7.74
6	Far	Good	349	66.49	-8.55
7	Near	Poor	925	64.46	-11.44
8	Moderate	Poor	490	63.99	-11.15
9	Far	Poor	158	63.41	-11.89

Overall Findings

- **Study Hours + Attendance** showed the strongest combined effect on academic performance.
- **Study Hours + Motivation** demonstrated that highly motivated students consistently outperformed students with the same study hours but lower motivation.
- **Parental Involvement** had a greater impact than **Parental Education Level**.
- **Access to Educational Resources** influenced performance more than **Family Income**.
- **Teacher Quality** consistently improved student performance and helped reduce the negative effects of **Low Family Income** and **Learning Disabilities**.
- **Positive Peer Influence** enhanced the effect of **High Motivation** on academic achievement.
- Students with **Low Previous Scores** benefited the most from increasing their **Study Hours**.
- **Attendance** proved to be a stronger predictor of academic success than **Distance from Home**.

Most Influential Combined Factors

1. Study Hours + Attendance
2. Study Hours + Motivation
3. Teacher Quality + Access to Resources
4. Parental Involvement + Parental Education
5. Family Income + Access to Resources
6. Teacher Quality + Family Income
7. Previous Scores + Study Hours
8. Peer Influence + Motivation
9. Learning Disabilities + Teacher Quality
10. Distance from Home + Attendance